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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCFIM100 - HEALTH AND FITNESS EDUCATION

(2024 ADMISSION ONWARDS)

Answer Key

Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. What is the primary goal of health promotion? [U] / [CO1]

   Ans: d) All of the above

2. Which of the following food groups should be included in a balanced diet? [U] / [CO4, CO5]

   Ans: a) Fruits, vegetables, whole grains, lean proteins, and healthy fats

3. What is the primary role of surveillance in vaccination programs? [U] / [CO2, CO3]

   Ans: c) Identifying and responding to vaccine-preventable disease outbreaks

4. What is the best way to incorporate rest into an exercise routine? [U] / [CO1, CO7]

   Ans: b) Schedule regular rest days

5. Health education helps in: [U] / [CO1]

   Ans: d) All of the above

6. What does CHA stand for? [U] / [CO3, CO6]

   Ans: a) Community Health Assessment

7. Why is environmental hygiene important in schools? [U] / [CO4]

   Ans: c) It helps prevent the spread of diseases

8. What is the first step in effective time management? [U] / [CO8]

   Ans: a) Making a to-do list

9. Deficiencies of which of the following nutrients can lead to anaemia? [U] / [CO4, CO5]

   Ans: b) Copper and iron

10. Which of the following is a common complication of uncontrolled hypertension? [U] / [CO2, CO4]

    Ans: d) All of the above

Part B

Short Answer Type Questions

Answer any **10** Questions

Each Question carries **2** marks

11. Discuss the future trends in school health services and their potential impact on students. [U] / [CO4, CO5, CO6]

Answer:

Future trends include digital health monitoring (apps, telehealth), mental-health counselling, nutrition programs, and regular fitness assessments. These trends improve early disease detection, promote healthy habits, and enhance students' physical, emotional, and academic well-being.

12. Examine the effects of poor nutrition on community health. What measures can be taken to promote healthy eating habits? [An] / [CO3, CO6]

Answer: Poor nutrition leads to obesity, anaemia, diabetes, weakened immunity, and increased healthcare burden. Measures include awareness programs, availability of healthy foods, school nutrition policies, food-labelling, and community campaigns promoting balanced diets

13. Identify and describe the key dimensions of health. [U] / [CO1]

Answer: The main dimensions are:

Physical health

Proper body functioning Mental health

Emotional stability social health

Healthy relationships Spiritual health

Values and purpose Environmental health

Safe surrounding

14. Define calories and why is it important in diet planning [U] / [CO4, CO5]

Answer: A calorie is a unit of energy obtained from food. It is important in diet planning because calorie intake determines energy balance, weight management, and supports body functions and physical activity

15. Discuss the importance of regular medical check-ups for students' overall health. [U] / [CO4, CO5, CO6]

Answer: Regular check-ups help early detection of health issues, monitor growth, ensure vaccination updates, identify vision/hearing problems, and promote preventive health behaviour

16 Evaluate the importance of skill-related fitness for athletes. [E] / [CO1]

. Answer:

Skill-related fitness components—agility, balance, coordination, power, speed, reaction time—

enhance athletic performance, reduce injury risk, and improve efficiency in sports-specific

movements

17 Describe the impact of non-communicable diseases on public health. [U] / [CO2, CO4]

. Answer: NCDs like diabetes, cancer, and hypertension increase disability, death rates, healthcare costs, and reduce productivity. They place long-term pressure on public health systems.

18 Discuss the impact of climate change on food security and nutrition in India. [U] / [CO4, CO5, CO6]

. Answer: Climate change affects rainfall, reduces crop yield, increases food prices, and lowers nutritional quality. It threatens access to safe, adequate food and increases malnutrition risk

19 Identify the role do recreational sports and physical activities in health [U] / [CO1, CO7]

. promotion? How can participation benefit individuals and communities?

Answer: Recreational sports promote fitness, reduce stress, prevent diseases, and improve mental well-being. For communities, they build teamwork, reduce social isolation, and encourage healthier lifestyles

20 Define Health and explain its significance [K] / [CO1]

. Answer: Health is a state of complete physical, mental, and social well-being. It is significant because it improves quality of life, productivity, longevity, and overall functioning

21 State the role of meditation in stress reduction? [K] / [CO8]

. Answer: Meditation lowers stress hormones, improves focus, calms the mind, reduces anxiety, and enhances emotional stability

22 Discuss the importance of accuracy and timeliness in health record [U] / [CO4, CO5, CO6]

. documentation.

Answer: Accurate and timely records ensure proper diagnosis, safe treatment, legal protection, continuity of care, and improved healthcare quality

[10x2 = 20]

Part C

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

23 Evaluate the importance of health-related physical fitness in today's society. [E] / [CO1]

· How do its components affect individual health, quality of life, and the prevention of chronic diseases.

Answer: Health-related physical fitness includes

cardiovascular endurance,

muscular strength,

flexibility,

muscular endurance,

and body composition.

It reduces the risk of chronic diseases like diabetes, hypertension, and obesity.

Improves heart and lung function, joint mobility, and muscle health. Enhances mental health by reducing stress and anxiety.

Promotes better sleep, immunity, and weight control.

Leads to better productivity and quality of life.

Physical fitness is essential for maintaining long-term health and preventing lifestyle-related illnesses

24 Discuss the importance of personal hygiene in maintaining overall health. [U] / [CO4]

Answer: Personal hygiene prevents infections, promotes social acceptance, improves self-confidence, and reduces disease transmission. Practices like handwashing, bathing, dental care, nail trimming, and clean clothing protect against communicable diseases. Good hygiene is vital for physical, mental, and social well-being and forms the base of public health

25. How can patient engagement be encouraged through the use of personal health records? [An] / [CO4, CO5, CO6]

Answer**: Patient engagement can be encouraged through Personal Health Records by:**

1. Providing easy access to health information – PHRs allow patients to view their medical history, lab results, and medications anytime, increasing involvement in their care.

2. Enhancing communication – Patients can message healthcare providers, ask questions, and receive reminders, improving participation.

3. Promoting self-management – Patients can track blood pressure, glucose levels, diet, and exercise, helping them take responsibility for their health.

4. Empowering informed decision-making – When patients understand their health conditions, they can make better lifestyle and treatment choices.

5. Increasing transparency – Access to records builds trust and motivates patients to stay engaged

26. Examine the impact of global health issues on local public health programs. How can local initiatives align with global health priorities? [An] / [CO3, CO6]

Answer**: Impact of global health issues on local programs:**

1. Emerging diseases like COVID-19 push local health systems to improve surveillance and emergency preparedness.

2. Global vaccination goals influence local immunization campaigns.

3. Climate change increases heat-related illnesses, vector-borne diseases, and pollution, requiring local adaptation strategies.

4. Global malnutrition and lifestyle diseases shape local nutrition and NCD (Non-communicable disease) programs.

5. International health regulations guide local reporting, monitoring, and response systems.

How local initiatives can align with global priorities:

1. Implement WHO guidelines for disease control, vaccination, and sanitation.

2. Adopt sustainable development goals (SDGs) such as good health, clean water, and sanitation.

3. Strengthen community-based surveillance to support global disease-tracking systems.

4. Partner with global agencies (WHO, UNICEF, NGOs) for funding and

technical support.

5. Promote awareness on global issues like antimicrobial resistance, climate health, and nutrition.

27. Define biological growth factors and discuss their significance in human development. How does regular physical fitness influence the production and effectiveness of these factors. [K] / [CO1]

Answer**: Biological growth factors:**

These are natural substances such as hormones, proteins, and peptides that regulate cell growth, division, repair, and overall development. Examples include growth hormone (GH), insulin-like growth factor (IGF-1), thyroid hormones, and sex hormones. Significance in human development:

1. Support tissue and organ growth in childhood and adolescence.

2. Promote bone growth and density, reducing fracture risk.

3. Facilitate muscle development and repair.

4. Regulate metabolism and energy use.

5. Support brain development, cognitive function, and emotional health.

Influence of physical fitness:

1. Increases growth hormone production, which enhances muscle and bone growth.

2. Boosts IGF-1 levels, improving tissue repair and cell regeneration.

3. Improves hormone balance, making growth factors work more effectively.

4. Strengthens muscles and bones, enhancing responsiveness to growth factors.

5. Reduces stress hormones, helping growth factors function optimally

28. Describe the importance of vaccination programs in preventing infectious diseases, highlighting public health strategies for increasing vaccination coverage. [U] / [CO2, CO3]

Answer: Importance of vaccination programs:

1. Prevents infectious diseases such as measles, polio, influenza, and COVID-19.

2. Reduces morbidity and mortality, especially in children and the elderly.

3. Achieves herd immunity, protecting vulnerable individuals who cannot be vaccinated.

4. Decreases healthcare costs by preventing outbreaks.

5. Eradicates or controls diseases globally (e.g., smallpox eradication, polio

elimination in many regions).

Public health strategies to increase vaccination coverage:

1. Awareness campaigns to educate about vaccine benefits and safety.
2. School-based vaccination programs to reach children efficiently.
3. Mobile clinics and outreach programs for rural and underserved populations.
4. Digital reminders and tracking systems to reduce missed doses.
5. Government policies such as mandatory vaccination for school admission.
6. Affordable or free vaccines through national immunization programs.
7. Training healthcare workers to address hesitancy and misinformation